# GOOD DIAGRAMS EXIST FOR FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 5277-5284, following the statement at 5257-5275.

*Editorial note.* No omissions.

*Allowed standard prerequisites (editorial).*

- Finite type closed in finite presentation. lemma-finite-type-closed-in-finite-presentation (same chapter, cited at ~5277-5279). Closed immersion X -> X' over B with X' fp over B.
- Fp morphisms descend to finite stages. lemma-descend-finite-presentation (same chapter, card L26286, cited at ~5280-5282). X' descends to X'_i -> B_i fp; set W = X'_i.

*Source reference.* Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0872: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3036–3036. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0CP6: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5249–5249. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *In Situation 0CP6. Let $X \to B$ be a quasi-separated and finite type morphism of algebraic spaces. Then there exists an $i \in I$ and a diagram*

$$\begin{array}{ccc} X & \longrightarrow & W \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_i \end{array} \tag{0.1.1}$$

*such that $W \to B_i$ is of finite type and such that the induced morphism $X \to B \times_{B_i} W$ is a closed immersion.*

*Date*: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* By Lemma 0872 we can find a closed immersion $X \to X'$ over $B$ where $X'$ is an algebraic space of finite presentation over $B$. By Lemma 07SK we can find an $i$ and a morphism of finite presentation $X'_i \to B_i$ whose pull back is $X'$. Set $W = X'_i$. $\square$